A LOCAL NON-CONTAINMENT IN STEP (V) OF IUT IV, THEOREM 1.10

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Abstract. Step (v) of the proof of Theorem 1.10 in the fourth paper of Mochizuki's series on inter-universal Teichmüller theory estimates, prime by prime, the log-volume of the holomorphic hull of the union of possible images of a Θ-pilot object defined in Corollary 3.12 of the third paper. For each tuple of places it uses an explicit comparison module, depending on the order of the q-parameter at the place carrying the label j. We give an explicit initial Θ-datum, with two places of multiplicative reduction above the prime 30965771, and a pure tensor obtained from a theta value by a label permutation whose image under any factorwise log-shell-preserving automorphism lies outside this module. Under the membership and orbit-closure readings stated in Section 4, an element of this form belongs to the projection of the possible-image union, and the corresponding hull component has larger log-volume than the displayed upper bound. The full identification with the source's intended construction remains unverified here; the example is offered as a concrete local consistency test and a basis for further discussion. No conclusion is drawn about the truth of Corollary 3.12 itself, the final global inequality of Theorem 1.10, or the abc conjecture.

1. The statement of Step (v)

We use the notation of [IUT1, IUT3, IUT4]; page numbers refer to the versions listed at the end, and we omit some decorations such as $^{n,\circ}$. For a finite extension k of $\mathbb{Q}_p$ we write $\mathcal{O}_k$ for its ring of integers, e_k for its ramification index, ord for the valuation on an algebraic closure of $\mathbb{Q}_p$ with $\mathrm{ord}(p) = 1$, and $\mathfrak{d}_k$ for the order of a generator of the different of $\mathcal{O}_k$ over $\mathbb{Z}_p$. For finitely many such fields $(k_i)_{i\in I}$ we write, as in [IUT4, Props. 1.1, 1.2, pp. 9–10],

$$k_I = \bigotimes_{i\in I} k_i, \qquad R_I = \bigotimes_{i\in I} \mathcal{O}_{k_i}, \qquad \log_p(R_I^\times) = \bigotimes_{i\in I} \log_p(\mathcal{O}_{k_i}^\times), \qquad \mathfrak{d}_I = \textstyle\sum_{i\in I} \mathfrak{d}_{k_i},$$

the tensor products being over $\mathbb{Q}_p$, respectively $\mathbb{Z}_p$, and $(R_I)^\sim$ for the normalization of R_I. The ring k_I is a finite product of fields L, and $(R_I)^\sim$ is the product of their rings of integers. We use the log-volume $\mu^{\log}$ of [IUT4, Prop. 1.4(i), p. 13], normalized by $\mu^{\log}((R_I)^\sim) = 0$ and $\mu^{\log}(p\cdot(R_I)^\sim) = -\log p$, and three elementary facts.

(1.1) In every factor L of k_I, the component of a pure tensor $y = \otimes_i z_i$, $z_i \neq 0$, has ord equal to $c = \sum_i \mathrm{ord}(z_i)$. Hence $(R_I)^\sim \cdot y = \{z \in k_I : \mathrm{ord}(z_L) \geq c \text{ for every } L\}$ and $\mu^{\log}((R_I)^\sim \cdot y) = -c\log p$.

(1.2) If p is odd and $e_k \leq p-2$, then $\log_p(\mathcal{O}_k^\times) = \{z \in k : \mathrm{ord}(z) \geq 1/e_k\}$ [IUT4, Prop. 1.2(i), p. 10]; so for $z \in k^\times$ and $n \in \mathbb{Z}$ we have $z \in p^n \log_p(\mathcal{O}_k^\times)$ if and only if $n < \mathrm{ord}(z)$.

(1.3) Write $\mathcal{A}(k)$ for the group of $\mathbb{Q}_p$-linear automorphisms of k that map $\log_p(\mathcal{O}_k^\times)$ onto itself, equivalently the log-shell $p^{-1}\log_p(\mathcal{O}_k^\times)$ (p odd) [IUT3, Rem. 1.2.2(i), p. 36]. Under the hypothesis of (1.2), every $\psi \in \mathcal{A}(k)$ maps each $p^n \log_p(\mathcal{O}_k^\times)$ onto itself, hence $\lceil \mathrm{ord}\,\psi(z)\rceil = \lceil \mathrm{ord}\, z\rceil$; in particular $\mathrm{ord}\,\psi(z) \leq \lceil \mathrm{ord}\, z\rceil$.

Fix an initial Θ-datum [IUT1, Def. 3.1, pp. 61–63] that satisfies the hypotheses of [IUT4, Thm. 1.10, pp. 22–23], a prime number p that ramifies in K, and $j \in \{1,\dots,\ell^*\}$, where $\ell^* = (\ell-1)/2$; write $\mathbb{S}^{\pm}_{j+1} = \{0,1,\dots,j\}$ [IUT4, p. 27]. For a place v of F_{mod} above p let $\underline{v} \in \underline{\mathbb{V}}$ be the corresponding element [IUT1, Def. 3.1(e)]; we write $k_v = K_{\underline{v}}$, $\underline{\underline{q}}_v = \underline{\underline{q}}_{\underline{v}}$ for the 2ℓ-th root of the q-parameter [IUT1, Ex. 3.2(iv), p. 71], and $\rho(v) = \mathrm{ord}_p(\underline{\underline{q}}_v)$. The packet $\mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1}; \mathcal{D}^{\vdash}_p)$ is the tensor product, over the labels in $\mathbb{S}^{\pm}_{j+1}$, of the direct sum over the places $v \mid p$ of the

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$\mathbb{Q}$-spans of the log-shells [IUT4, p. 27], [IUT3, Prop. 3.2, pp. 97–98]. Identifying the $\mathbb{Q}$-span of the log-shell at v with k_v [IUT3, Rem. 3.1.1(i), p. 93; Prop. 3.2(i)], we get

$$\mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1};\mathcal{D}^{\vdash}_{p}) = \bigoplus_{\mathbf{v}} k_{\mathbf{v}}, \qquad k_{\mathbf{v}} = k_{v_0} \otimes k_{v_1} \otimes \cdots \otimes k_{v_j},$$

the sum running over the tuples $\mathbf{v} = (v_0, \dots, v_j)$ of places of F_{mod} above p; the factor of $k_{\mathbf{v}}$ with label i is k_{v_i}. Write $\mathrm{pr}_{\mathbf{v}}$ for the projection to $k_{\mathbf{v}}$, U_j for the union of the possible images of a Θ-pilot object, subject to the indeterminacies (Ind1), (Ind2), (Ind3), that is formed in the proof of [IUT3, Cor. 3.12] (denoted there $\mathcal{U}_{j,v_{\mathbb{Q}}}$, with $v_{\mathbb{Q}} = p$; p. 174), and $\overline{U}_j$ for its holomorphic hull [IUT3, Rem. 3.9.5(i), p. 127].

For a tuple $\mathbf{v}$, Step (v) [IUT4, pp. 27–28] applies [IUT4, Prop. 1.4(iii), p. 13] with $I = \mathbb{S}^{\pm}_{j+1}$, $k_i = k_{v_i}$, $I^* = \{i : e_{k_{v_i}} > p-2\}$, $i^{\dagger} = j$, and $\lambda = \mathrm{ord}(\underline{\underline{q}}^{j^2}_{v_j}) = j^2\rho(v_j)$ if $\underline{v}_j \in \underline{\mathbb{V}}^{\mathrm{bad}}$ ($\lambda = 0$ otherwise). From the inclusion $\phi(p^{\lambda}\cdot(R_I)^{\sim}) \subseteq C_{\mathbf{v}}$ of that proposition, where

$$C_{\mathbf{v}} = p^{\lfloor \lambda - \mathfrak{d}_I - a_I \rfloor}\cdot \log_p(R_I^{\times}) \ \subseteq \ M_{\mathbf{v}} = p^{\lfloor \lambda - \mathfrak{d}_I - a_I \rfloor - b_I}\cdot (R_I)^{\sim}$$

and a_I, b_I are as in [IUT4, Prop. 1.2], it concludes that a suitable multiple of $\log_p(R_I^{\times})$ contains the union of possible images,[1] and that an upper bound for the component at $\mathbf{v}$ of the log-volume of the holomorphic hull may be obtained from an upper bound for the log-volume of $M_{\mathbf{v}}$ [IUT4, p. 28]. If $I^* = \emptyset$, the latter is

$$\mu^{\log}(M_{\mathbf{v}}) \le (-\lambda + \mathfrak{d}_I + 1)\cdot \log p \tag{1}$$

[IUT4, Prop. 1.4(iii)]. This is the bound that we test.

Lemma 1. *The union of possible images whose holomorphic hull Step (v) estimates is U_j, and its component at the tuple $\mathbf{v}$ is $\mathrm{pr}_{\mathbf{v}}(U_j)$.*

Proof. Step (v) states that the container "contains the 'union of possible images of a Θ-pilot object' discussed in Step (iv)" [IUT4, p. 27]. Step (iv) introduces this union "according to the various definitions given in the statement of [IUTchIII], Corollary 3.12" [IUT4, p. 26], and estimates, for fixed $v_{\mathbb{Q}}$ and j, its component in $\mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1};\mathcal{D}^{\vdash}_{v_{\mathbb{Q}}})$ and then the component corresponding to each tuple $\{v_i\}$ [IUT4, p. 27]. In the proof of [IUT3, Cor. 3.12] this union is the subset $\mathcal{U}_{j,v_{\mathbb{Q}}} \subseteq \mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1};\mathcal{D}^{\vdash}_{v_{\mathbb{Q}}})$, formed before its holomorphic hull is taken [IUT3, p. 174]. For $v_{\mathbb{Q}} = p$, its component at the tuple $\mathbf{v}$ is $\mathrm{pr}_{\mathbf{v}}(U_j)$. □

2. The datum D_2

Let $F_0 = \mathbb{Q}(\sqrt{5})$, $\ell = 7$, and

$$\pi = 14164 + 5825\sqrt{5}, \qquad p = \mathrm{N}_{F_0/\mathbb{Q}}(\pi) = 30965771, \qquad a = \pi\bar{\pi}^{\,6},$$

where the bar denotes the conjugation of F_0. Let $E_0\colon y^2 = x(x-1)(x-a)$ over F_0, $F = F_0(\sqrt{-1}, E_0[15])$, $K = F(E_0[7])$, X_F the curve $E_0 \times_{F_0} F$ minus its origin, and $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ the set of places v of F_0 with $v \nmid 14$ and $\mathrm{ord}_v(j(E_0)) < 0$.

Proposition 2. *These data extend to an initial Θ-datum $D_2 = (\overline{F}/F, X_F, \ell, \underline{C}_K, \underline{\mathbb{V}}, \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}, \underline{\epsilon})$ in the sense of* [IUT1, Def. 3.1] *that satisfies the hypotheses of* [IUT4, Thm. 1.10]*, with $F_{\mathrm{mod}} = F_0$. The number p is prime and splits in F_0; the two places $b_1 = v_{\pi}$ and $b_2 = v_{\bar{\pi}}$ above p lie in $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$, have $[(F_{\mathrm{mod}})_{b_i} : \mathbb{Q}_p] = 1$, and have the local data of Table 1. Moreover $p > e^{*}_{\mathrm{mod}}\cdot\ell = 7741440$.*

Proof of Proposition 2, except Table 1. The integer p is prime, $p \equiv 1 \pmod 5$ and $p \equiv 3 \pmod 4$. So $p = \pi\bar{\pi}$ splits in F_0, both places above it have $[(F_0)_{b_i} : \mathbb{Q}_p] = 1$, and $\mathbb{Q}_p(\sqrt{-1})$ is unramified over $\mathbb{Q}_p$.

[1] In the words of [IUT4, p. 27], the inclusion "implies that the result of multiplying $p^{\lfloor\lambda\rfloor - |I|}\cdot 2^{-|I|}\cdot \log_p(R_I^{\times})$ by a suitable nonpositive [...] integer power of $p_{v_{\mathbb{Q}}}$ contains the 'union of possible images of a Θ-pilot object' discussed in Step (iv)". Which power is meant plays no role below: Theorem 5 concerns $M_{\mathbf{v}}$ and the bound (1).

place v	$\mathrm{ord}_v(a)$	$\mathrm{ord}_p(q_v)$	e_{k_v}	different exponent	$\mathfrak{d}_{k_v}$	$\rho(v)$
b_1	1	2	105	104	104/105	1/7
b_2	6	12	35	34	34/35	6/7

TABLE 1. The places of D_2 above $p = 30965771$; $k_v = K_{\underline{v}}$ and $\rho(v) = \mathrm{ord}_p(\underline{\underline{q}}_v)$.

The integer $\mathrm{N}(1-a)$. We have $\mathrm{N}(1-a) = p^7 - p\,\mathrm{Tr}(\pi^5) + 1 = 2^2 \cdot 11^2 \cdot R$, where

$$R = 564063402201437105083137489340735509737726645920 51$$

is prime. Hence no prime divides $\mathrm{N}(1-a)$ to the seventh power, $\mathrm{ord}_v(1-a) \le 6$ at every place v of F_0, and $\mathrm{ord}_2(1-a) = 1$ at the place above 2, which is inert. Neither $\mathrm{N}(a) = p^7$ nor $\mathrm{N}(1-a)$ is divisible by 7.

The curve. We have $j = j(E_0) = 2^8(a^2-a+1)^3/(a^2(a-1)^2)$ and $a \in \mathcal{O}_{F_0}$. At an odd place v with $\mathrm{ord}_v(a) > 0$ or $\mathrm{ord}_v(1-a) > 0$ the element $a^2 - a + 1$ is a unit, so $\mathrm{ord}_v(j) = -2\,\mathrm{ord}_v(a)$, respectively $-2\,\mathrm{ord}_v(1-a)$; at the other odd places the discriminant $16a^2(a-1)^2$ is a unit. At b_1 and b_2, $1-a$ is a unit, so $\mathrm{ord}_{b_1}(j) = -2$ and $\mathrm{ord}_{b_2}(j) = -12$. A rational j would have equal orders at the two places above p; hence $j \notin \mathbb{Q}$ and $F_{\mathrm{mod}} = \mathbb{Q}(j) = F_0$ [IUT4, Prop. 1.8(ii), p. 19]. At the place above 2, a is a unit and $\mathrm{ord}_2(j) = 8 - 2 = 6 \ge 0$.

Conditions (a), (b) of [IUT1, Def. 3.1]. We have $\sqrt{-1} \in F$. As $E_0[2]$ is rational over F_0, $F = F_0(\sqrt{-1}, E_0[30])$; it is Galois over F_0, its Galois group embeds in $\mathbb{Z}/2 \times \mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})$, whose order is $2^{10} \cdot 3^2 \cdot 5$, so $[F : F_0]$ is prime to 7; and $E_F[6]$ is rational over F. By [IUT4, Prop. 1.8(v), p. 19], applied with the primes 3 and 5, E_F has semistable reduction at every finite place of F; so its reduction is multiplicative where $\mathrm{ord}(j) < 0$ and good where $\mathrm{ord}(j) \ge 0$ [Sil, Props. VII.5.1, VII.5.5]. By the previous paragraph, the places of multiplicative reduction are exactly the places above $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$; this set is nonempty ($b_1, b_2 \in \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$), its residue characteristics are odd, and E_F has good reduction at every other place not dividing $2 \cdot 7$, which is the reduction hypothesis of [IUT4, Thm. 1.10]. The field F has the form required there: $F_{\mathrm{tpd}} = F_{\mathrm{mod}}(E[2]) = F_0$, E_0 is in Legendre form, and $F = F_{\mathrm{tpd}}(\sqrt{-1}, E_0[3 \cdot 5])$.

Condition (c). The prime $\ell = 7$ is not a residue characteristic of $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$. At a place w of F above $v \in \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ the q-parameter has order $-\mathrm{ord}_w(j) = 2\,e(w|v)\,r$, where $r = \mathrm{ord}_v(a)$ or $\mathrm{ord}_v(1-a)$ lies in $\{1, \dots, 6\}$ (a vanishes only at b_1 and b_2, to the orders 1 and 6) and $e(w|v)$ divides $[F : F_0]$; so these orders are prime to 7. At such a place w the Legendre equation is minimal, as $c_4 = 16(a^2 - a + 1)$ is a unit [Sil, §VII.1], and its reduction has a node, at $(0,0)$ with tangents $y = \pm\sqrt{-1}\,x$ if $a \equiv 0$ and at $(1,0)$ with tangents $y = \pm(x-1)$ if $a \equiv 1$, rational over the residue field; so E_F has split multiplicative reduction and $E_F \cong E_q$ over F_w [Sil, App. C, Thm. 14.1]. For the image of G_F in $\mathrm{GL}(E_0[7]) \cong \mathrm{GL}_2(\mathbb{F}_7)$: above b_1, $a \equiv 0$ and $\mathrm{ord}_w(q) = 2e(w|b_1)$ is prime to 7. As $q^{1/7}$ generates a ramified extension of the maximal unramified extension of F_w, which contains μ_7, an element of inertia acts on $E_0[7] \cong \langle \zeta_7, q^{1/7} \rangle$ by a nontrivial transvection t. At the place of F_0 of degree one above 19 at which $\sqrt{5} \equiv 9$, E_0 has good reduction, $a \equiv 15$ and $a_{19} = 19 + 1 - \#\tilde{E}_0(\mathbb{F}_{19}) = -4$; the Frobenius g acts on $E_0[7] \cong \tilde{E}_0[7]$ [Sil, Prop. VII.3.1] and satisfies $g^2 + 4g + 19 = 0$ [Sil, Thm. V.2.3.1]; as $X^2 + 4X + 5$ is irreducible over $\mathbb{F}_7$ (its discriminant 3 is not a square), g fixes no line. The image H of G_F is normal in the image of G_{F_0}, so it contains t and gtg^{-1}, transvections with distinct fixed lines, which generate $\mathrm{SL}_2(\mathbb{F}_7)$.

Conditions (d)–(f). We do not invoke [IUT4, Cor. 2.2] for D_2: its condition (P1) [IUT4, p. 45] contains the inequality $h^{1/2} \le \ell$ for the height h defined in its proof (p. 44), and here $h > \ell^2$. We construct the data directly; as noted on p. 46, the essential input for $\underline{\mathbb{V}}$ and $\underline{\epsilon}$ is the property, shown above, that the image of G_F contains $\mathrm{SL}_2(\mathbb{F}_\ell)$. Since $j \notin \mathbb{Q}$, j is none of $2^{14} \cdot 31^3 \cdot 5^{-3}$, $2^2 \cdot 73^3 \cdot 3^{-4}$, 1728, 0, so X_F admits an F-core [MFHMP, Prop. 2.1]. Hence $X_{\overline{F}}$ is not arithmetic [CanLift, Prop. 2.3(i)], nor is its finite étale quotient $C_{\overline{F}} = [X_{\overline{F}}/\{\pm 1\}]$, a punctured hemi-elliptic orbicurve, which is therefore an $\overline{F}$-core [CanLift, Prop. 2.7]; so $C_F = [X_F/\{\pm 1\}]$ and $C_K = C_F \times_F K$ are an F-core and a K-core [CanLift, Prop. 2.3(i)]. The group $E_0[7]$ is rational

over K. Fix a line $W \subset E_0[7]$ and $0 \neq \xi \in E_0[7]/W$; let $\alpha\colon E_K \to E' = E_K/W$ be the quotient and $\beta\colon E' \to E_K$ the isogeny with $\beta\alpha = [7]$. The finite étale cover $\beta\colon \underline{X}_K = E' \setminus \ker\beta \to X_K$ has group $\ker\beta = \alpha(E_0[7]) \cong E_0[7]/W$, corresponds to the quotient $E_0[7] \to E_0[7]/W$ of the maximal abelian quotient modulo 7 of the geometric fundamental group of X_K, and has its cusps, the points of $\ker\beta$, rational over K; so $\underline{C}_K = [\underline{X}_K/\{\pm 1\}]$, formed with the cusp 0 as origin, is of type $(1, 7\text{-tors})^{\pm}$ [EtTh, Def. 2.1], and, being finite étale over the K-core C_K, it has K-core C_K, as required in (d). Let $\underline{\epsilon}$ be the cusp of $\underline{C}_K$ given by $\pm\xi$, as in (f). At a place w of K above an element of $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$, the Tate uniformization of E_0 over K_w, which exists by the paragraph on (c), gives a line $\mu_7 \subset E_0[7]$ and a generator ± 1, canonical up to sign, of $E_0[7]/\mu_7 \cong \mathbb{Z}/7\mathbb{Z}$; if $(W, \pm\xi)$ is this pair, then $\underline{C}_K$ is of type $(1, \mathbb{Z}/7\mathbb{Z})^{\pm}$ at w [EtTh, Def. 2.5(i)] and $\underline{\epsilon}_w$ is the cusp of ± 1, as required in (e), (f). The group $\mathrm{SL}_2(\mathbb{F}_7)$ acts transitively on the pairs $(L, \pm\eta)$ of a line $L \subset E_0[7]$ and $0 \neq \eta \in E_0[7]/L$ (it sends any basis of determinant 1 to any other), and it lies in the image of $\mathrm{Gal}(K/F)$. Since E_0 and its Tate parameter are defined over the completion of F below w, an element $\gamma \in \mathrm{Gal}(K/F)$ carries the pair at w to the pair at $\gamma(w)$, and we may choose γ so that the latter is $(W, \pm\xi)$. Choosing such a place over each element of $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$, and any place over the others, gives $\underline{\mathbb{V}}$.

The threshold. $F_{\mathrm{mod}} = \mathbb{Q}(\sqrt{5})$ has $e_{\mathrm{mod}} = 2$, so $e^*_{\mathrm{mod}} \cdot \ell = 2^{12} \cdot 3^3 \cdot 5 \cdot 2 \cdot 7 = 7741440 < p$ [IUT4, p. 22]. □

3. The places above p

Proof of Table 1. Let $b \in \{b_1, b_2\}$ and $m = -\mathrm{ord}_b(j)$, so $m = 2$ at b_1 and $m = 12$ at b_2; let $k_0 = (F_0)_b(\sqrt{-1}) = \mathbb{Q}_p(\sqrt{-1})$. As in the previous proof, E_0 has split multiplicative reduction over k_0, so $E_0 \cong E_q$ over k_0 with $\mathrm{ord}_p(q) = m$, and for n prime to p the field $k_0(E_0[n])$ is $k_0(\mu_n, q^{1/n})$ [Sil, App. C, Thm. 14.1, and the expansion $j = q^{-1} + 744 + \cdots$ on p. 444]. Write $q = u\,p^m$ with $u \in \mathcal{O}^{\times}_{k_0}$. The field $k_0(\mu_n, u^{1/n})$ is unramified over $\mathbb{Q}_p$, and adjoining to it an n-th root of p^m gives a totally ramified extension of degree $n/\gcd(n, m)$. Hence $k_0(E_0[n])$ is tamely ramified over $\mathbb{Q}_p$, with ramification index $n/\gcd(n, m)$. For $\underline{v} \in \underline{\mathbb{V}}$ above b we have $k_v = K_{\underline{v}} = k_0(E_0[105])$, so $e_{k_v} = 105/\gcd(105, m)$, which is 105 at b_1 and 35 at b_2. Since k_v is tame over $\mathbb{Q}_p$, its different has exponent $e_{k_v} - 1$, that is $\mathfrak{d}_{k_v} = (e_{k_v} - 1)/e_{k_v}$ [IUT4, Prop. 1.3(i), pp. 11–12]. Finally $\underline{\underline{q}}_v = q^{1/2\ell}$ [IUT1, Ex. 3.2(iv), p. 71], so $\rho(v) = m/14$, which is $1/7$ at b_1 and $6/7$ at b_2. Both e_{k_v} are at most $p - 2$. □

The integer calculations of §§2–3 were independently checked by two implementations.

4. Two elements of the union

Readings and scope. This note presents an explicit local non-containment calculation. Its application to U_j is conditional on (M) and (R), including the compatibility with the cited constructions of the local-coordinate descriptions used in Lemmas 3 and 4. A complete identification with the source construction is not established in this note.

- (M) At a selected bad place, the local region of the Θ-pilot is $\theta\Lambda$, where θ is the relevant splitting-monoid generator and the unit lattice Λ contains $1 \otimes \cdots \otimes 1$; its image under the relevant Kummer comparison of [IUT3, Thm. 3.11(ii)] is counted among the possible-image regions.
- (R) The family of possible-image regions is stable under the packet maps induced by the procession automorphisms defining (Ind1), including the label permutations used below, and under (Ind2).

These are readings of [IUT3, Prop. 3.7(v), Def. 3.8(i), Thm. 3.11, Cor. 3.12], not a new definition of U_j. The local non-containment calculation in Theorem 5 does not require these readings; its application to U_j does.

Lemma 3. *Assume* (M). *Let* $\mathbf{v}$ *be a tuple with* $\underline{v}_j \in \underline{\mathbb{V}}^{\mathrm{bad}}$, *and let* $\theta \in k_{v_j}$ *be a generator of the component labelled* j *at* $\underline{v}_j$ *of the splitting monoid, that is, the theta value* $\underline{\underline{q}}_{v_j}^{j^2}$ *up to a root of unity* [IUT3, p. 11]; *so* $\mathrm{ord}(\theta) = j^2\rho(v_j)$. *Let* $x \in k_{\mathbf{v}}$ *be the pure tensor with* θ *at the factor*

of label j and 1 at the other factors. Then U_j contains an element X with $\mathrm{pr}_{\mathbf{v}}(X) = \chi(x)$ for some $\chi = \otimes_i \chi_i$, $\chi_i \in \mathcal{A}(k_{v_i})$.

Proof. The Θ-pilot object is determined by generators, at the places of $\underline{\mathbb{V}}^{\mathrm{bad}}$, of the monoids of [IUT3, Def. 3.8(i), p. 112], and it is constructed from the local fractional ideals generated by elements of these monoids [IUT3, Prop. 3.7(v), p. 112]. At $\underline{v} \in \underline{\mathbb{V}}^{\mathrm{bad}}$ the component labelled j of these monoids is a subset of the part of the packet whose factor of label j is the summand at $\underline{v}$, and it acts on this part multiplicatively [IUT3, Prop. 3.4(ii), pp. 102–103]; it lies there through the natural homomorphism of [IUT3, Prop. 3.4(i)], that is, by tensoring with 1 in the factors of the other labels [IUT3, Prop. 3.1(ii), p. 93]. A fractional ideal generated by an element contains that element. The possible images of [IUT3, Cor. 3.12] are the images of the Θ-pilot object relative to the Kummer isomorphisms of [IUT3, Thm. 3.11(ii), p. 155], which include isomorphisms of these splitting monoids and of the parts of the packets just described, in the multiradial representation of [IUT3, Thm. 3.11(i)], subject to (Ind1), (Ind2), (Ind3), and U_j is their union [IUT3, pp. 173–174]. Let g be the element with θ in the factor of label j and 1 in the other factors, in the packet of the Hodge theater in which the Θ-pilot object is formed. By (M) it lies in the corresponding region of the Θ-pilot object, and its Kummer image is counted among the possible images; hence U_j contains $X = \kappa(g)$, where κ is the Kummer isomorphism of packets onto $\mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1}; \mathcal{D}^{\vdash}_p)$ of [IUT3, Thm. 3.11(ii)(a)], built place by place from isomorphisms of log-shells [IUT3, Prop. 3.2]. In that Hodge theater the $\mathbb{Q}$-spans of the log-shells are the fields k_v themselves, and the component of g at $\mathbf{v}$ is x. For $\mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1}; \mathcal{D}^{\vdash}_p)$ the identifications of the $\mathbb{Q}$-spans of the log-shells with the fields k_v, used in §1, form an Ism-orbit compatible with the log-shells [IUT3, Prop. 1.2(vi), p. 32]. Read through them, κ maps each summand $k_{\mathbf{w}}$ to itself by a tensor product of automorphisms that preserve the log-shells, so

$$\mathrm{pr}_{\mathbf{v}}(X) = \mathrm{pr}_{\mathbf{v}}(\kappa(g)) = \chi\big(\mathrm{pr}_{\mathbf{v}}(g)\big) = \chi(x), \qquad \chi = \otimes_i \chi_i, \quad \chi_i \in \mathcal{A}(k_{v_i}).$$

□

Choice of theta representative. Under the small-ramification assumption in (1.2), the local non-containment estimate is independent of the choice within the $\mu_{2\ell}$-orbit of the theta value [IUT3, Rem. 3.11.4(i), p. 171], with the local field fixed. Indeed, if $\zeta \in \mu_{2\ell}(k_{v_j})$, then multiplication by ζ is $\mathbb{Q}_p$-linear and preserves $\log_p(\mathcal{O}^{\times}_{k_{v_j}})$, by (1.2) and $\mathrm{ord}(\zeta) = 0$. It can therefore be absorbed into χ_j. This does not require χ_j to fix 1 or to preserve multiplication, and it does not establish membership in U_j independently of (M).

For a permutation σ of $\mathbb{S}^{\pm}_{j+1}$ and a tuple $\mathbf{v}$, let $\sigma_*\colon k_{\mathbf{v}} \to k_{\mathbf{v}\circ\sigma^{-1}}$ send the factor of label i to the factor of label $\sigma(i)$: $\sigma_*(\otimes_i z_i) = \otimes_i z_{\sigma^{-1}(i)}$.

Lemma 4. *Assume* (R). *Let σ be a permutation of $\mathbb{S}^{\pm}_{j+1}$ with $\sigma(0) = 0$, let $X \in U_j$, let $\mathbf{v}$ be a tuple, and put $\mathbf{w} = \mathbf{v} \circ \sigma^{-1}$. The images of X under the indeterminacy (Ind1) attached to σ, followed by (Ind2), exist and lie in U_j, and the component at $\mathbf{w}$ of each of them has the form $\psi(\sigma_*(\mathrm{pr}_{\mathbf{v}} X))$ with $\psi = \otimes_i \psi_i$, $\psi_i \in \mathcal{A}(k_{w_i})$. In particular $\psi(\sigma_*(\mathrm{pr}_{\mathbf{v}} X)) \in \mathrm{pr}_{\mathbf{w}}(U_j)$.*

Proof. A morphism of processions consists of an order-preserving injection of the stages and, at each stage, a capsule-full poly-morphism, with no condition relating the stages [IUT1, Def. 4.10, p. 120]; a capsule-full poly-isomorphism over a bijection of index sets is the set of all collections of isomorphisms between the corresponding members [IUT1, §0, p. 34]. So σ at the stage with index set $\mathbb{S}^{\pm}_{j+1}$, together with the identity at the other stages, is an automorphism of the procession of [IUT3, Thm. 3.11(i)], whose index sets the source regards as "subject to arbitrary permutation automorphisms" [IUT3, p. 10]; cf. [IUT1, Prop. 6.9(i), p. 169]. (Ind1) consists of the indeterminacies induced by such automorphisms, and (Ind2) of the action of independent copies of Ism on the direct summands of the factors [IUT3, Thm. 3.11(i), p. 154]. By (R), their images of possible-image regions are again possible-image regions. The packet is a tensor product over the labels of direct sums over the places [IUT3, Prop. 3.2], and a morphism of $\mathcal{D}^{\vdash}$-prime-strips is a collection of morphisms indexed by the places [IUT1, Def. 4.1(iv), p. 96]; so a member of the poly-isomorphism over σ maps each summand $k_{\mathbf{v}}$ to $k_{\mathbf{v}\circ\sigma^{-1}}$, sending the factor

of label i to that of label $\sigma(i)$ by the map induced by its component at i, and the component at $\mathbf{w}$ of the image of X is the image of $\mathrm{pr}_{\mathbf{v}}(X)$. Such members exist, the members of the capsule being isomorphic [IUT1, Def. 4.1(iii)]. The log-shells are constructed functorially from the prime-strips, and the identifications of their $\mathbb{Q}$-spans with the fields k_v form an Ism-orbit compatible with the log-shells [IUT3, Prop. 1.2(vi), p. 32]. Hence each member, followed by any element of (Ind2), maps $\mathrm{pr}_{\mathbf{v}}(X)$ to $\psi(\sigma_*(\mathrm{pr}_{\mathbf{v}}\, X))$ with $\psi_i \in \mathcal{A}(k_{w_i})$; that is, the images of X are $R(X)$ for maps R of the packet with $\mathrm{pr}_{\mathbf{w}} \circ R = \psi \circ \sigma_* \circ \mathrm{pr}_{\mathbf{v}}$. □

Related expositions. Hoshi [HoshiII, §25, p. 93 of the author's version] describes the hull of the union of the orbit of the Θ-pilot under the indeterminacies. After fixing identifications, Dupuy–Hilado [DH1, §4.6, pp. 19–20] describe procession automorphisms by permuting tensor factors and applying factorwise automorphisms, with target the summand indexed by the permuted tuple. These descriptions support the reading used here, but do not independently verify this example; see also the qualification concerning their formulas in [DH1, Rem. 4.10.1].

5. The non-containment

Let D_2 be as in Proposition 2 and $j = 3$, so $\mathbb{S}_4^{\pm} = \{0, 1, 2, 3\}$. Let $\mathbf{u} = (b_1, b_1, b_2, b_1)$, whose factor of label 3 lies over b_1, and take the local theta value $\theta = \underline{\underline{q}}_{b_1}^{9}$ up to a root of unity, with $\mathrm{ord}(\theta) = \frac{9}{7}$. Put $x = 1 \otimes 1 \otimes 1 \otimes \theta \in k_{\mathbf{u}}$. For $\sigma = (2\,3)$ put $\mathbf{v} = \mathbf{u} \circ \sigma^{-1} = (b_1, b_1, b_1, b_2)$ and

$$y = \sigma_*(x) = 1 \otimes 1 \otimes \theta \otimes 1 \;\in\; k_{b_1} \otimes k_{b_1} \otimes k_{b_1} \otimes k_{b_2}.$$

Under (M), Lemma 3 gives $X \in U_3$ with $\mathrm{pr}_{\mathbf{u}}(X) = \chi(x)$. Under (R), Lemma 4 then gives an element $\psi(\sigma_*(\chi(x))) = \psi'(y)$ of $\mathrm{pr}_{\mathbf{v}}(U_3)$, where $\psi' = \otimes_i \psi'_i$ with $\psi'_i = \psi_i \circ \chi_{\sigma^{-1}(i)} \in \mathcal{A}(k_{v_i})$, since $\sigma_* \circ \chi = (\otimes_i \chi_{\sigma^{-1}(i)}) \circ \sigma_*$.

Theorem 5. *For the initial Θ-datum D_2, $j = 3$ and $\mathbf{v} = (b_1, b_1, b_1, b_2)$, no element $\psi(y)$ with $\psi = \otimes_i \psi_i$, $\psi_i \in \mathcal{A}(k_{v_i})$, lies in the module $M_{\mathbf{v}}$ of §1. Under the readings* (M) *and* (R) *of §4, it follows that* $\mathrm{pr}_{\mathbf{v}}(U_3) \not\subseteq M_{\mathbf{v}}$ *and that the component* $\mathrm{pr}_{\mathbf{v}}(\overline{U}_3)$ *of the holomorphic hull has log-volume at least* $-2\log p$ *(possibly* $+\infty$*), whereas the displayed upper bound* (1) *is* $-\frac{97}{35}\log p$.

Proof. Step (v) takes $i^{\dagger} = 3$, whose factor lies over b_2, so by Table 1

$$\lambda = 9 \cdot \tfrac{6}{7} = \tfrac{54}{7}, \qquad \mathfrak{d}_I = 3 \cdot \tfrac{104}{105} + \tfrac{34}{35} = \tfrac{138}{35}, \qquad I^* = \emptyset,$$

and (1) gives $\mu^{\log}(M_{\mathbf{v}}) \le (-\frac{54}{7} + \frac{138}{35} + 1)\log p = -\frac{97}{35}\log p$. By (1.3), $\mathrm{ord}\,\psi_2(\theta) \le \lceil \frac{9}{7} \rceil = 2$ and $\mathrm{ord}\,\psi_i(1) \le 0$ for $i \neq 2$; so by (1.1) every component of $\psi(y)$ has ord ≤ 2, and $(R_I)^{\sim} \cdot \psi(y) \supseteq p^2 \cdot (R_I)^{\sim}$. If $\psi(y) \in M_{\mathbf{v}}$, then, $M_{\mathbf{v}}$ being an $(R_I)^{\sim}$-module,

$$-\tfrac{97}{35}\log p \;\ge\; \mu^{\log}(M_{\mathbf{v}}) \;\ge\; \mu^{\log}(p^2 \cdot (R_I)^{\sim}) = -2\log p,$$

which is false: $-\frac{97}{35} < -2$, with margin $\frac{27}{35}\log p$. Under (M) and (R), Lemmas 3 and 4 show that $\mathrm{pr}_{\mathbf{v}}(U_3)$ contains an element of this form, so it is not contained in $M_{\mathbf{v}}$. For the hull: if U_3 is not relatively compact, its holomorphic hull is the whole packet [IUT3, Rem. 3.9.5(i)], of log-volume $+\infty$. Otherwise the hull is $\eta \cdot \mathcal{O}$ for an element η with nonzero components [IUT3, ibid.], so $\mathrm{pr}_{\mathbf{v}}(\overline{U}_3)$ is an $(R_I)^{\sim}$-module containing $\psi(y)$, hence $p^2 \cdot (R_I)^{\sim}$, and its log-volume is at least $\mu^{\log}(p^2 \cdot (R_I)^{\sim}) = -2\log p$. □

Exact volume of the comparison module. For the same datum, [IUT4, Props. 1.2, 1.4(i)] give

$$a_I = \frac{3}{105} + \frac{1}{35} = \frac{2}{35}, \qquad b_I = -\frac{2}{35}, \qquad \lfloor \lambda - \mathfrak{d}_I - a_I \rfloor = \left\lfloor \frac{26}{7} \right\rfloor = 3.$$

Hence, in the fractional-ideal notation of §1,

$$M_{\mathbf{v}} = p^{107/35}(R_I)^{\sim}, \qquad \mu^{\log}(M_{\mathbf{v}}) = -\frac{107}{35}\log p.$$

The difference between the lower bound $-2\log p$ for any hull containing such an image and this exact module volume is $\frac{37}{35}\log p$. This refines the comparison with the module itself; the difference from the displayed upper bound (1) remains $\frac{27}{35}\log p$.

Thus, under (M) and (R), the projected possible-image union and its holomorphic hull are incompatible with the component bound attributed to Step (v). Dupuy–Hilado [DH, §6.2, footnote 7] note that their simplified example for estimating the hull in §6.2 of their paper does not include the permutations between tensor factors present in the actual (Ind1), so the computation in that subsection cannot be applied directly.

The example above concerns the local containment and the local upper bound used in Step (v) for a single component. It does not by itself decide the global inequality obtained later in Theorem 1.10 after summing over all places. No assertion is made here concerning Corollary 3.12 itself or the abc conjecture.

Use of AI tools. Generative AI tools (large language models: Claude by Anthropic, and ChatGPT and Codex by OpenAI) were used extensively in this work, under the author's direction, in finding the example and its proofs, in the computations and in the writing. The author takes full responsibility for the contents of this note.

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